

Ayush Agarwal

Live Project: `lexai-3fd1a.web.app`
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EDUCATION

- VIT Bhopal University** Bhopal, India
 - Bachelor of Technology - Computer Science and Engineering; CGPA: 8.50/10* *Sep 2023 – Ongoing*
 - Courses: Data Structures, Algorithms, Machine Learning, Deep Learning, NLP, Databases, Cloud Computing*

SKILLS SUMMARY

- Programming:** Python
- Backend:** FastAPI, REST APIs, WebSockets, Async Programming, SQLAlchemy
- AI/ML:** LLMs, RAG, NLP, Prompt Engineering, Multi-Agent Systems, LangChain, MCP, Tool Calling
- Databases:** SQLite (FTS5), PostgreSQL, Firestore, Azure Cosmos DB
- Cloud:** Google Cloud Run, Cloud Build, Firebase Hosting, Microsoft Azure, Docker
- Retrieval:** FAISS, BM25, Vector Search, PageRank, Reciprocal Rank Fusion
- Libraries:** Scikit-learn, Pandas, NumPy, Streamlit

EXPERIENCE

- Docu3C Technologies** Remote (Seattle, Washington, USA)
 - AI Software Engineer Intern* *Aug 2025 – July 2026*
 - Backend Engineering:** Developed asynchronous FastAPI services using REST APIs, Server-Sent Events (SSE), and WebSockets to support concurrent AI inference, streaming responses, and scalable document processing workflows.
 - Agentic AI Systems:** Designed and implemented multi-agent AI workflows using Model Context Protocol (MCP) with tool-calling capabilities, reducing manual financial document review by 60% across production document intelligence pipelines.
 - LLM Engineering:** Built production-grade LLM pipelines with structured outputs using Pydantic and Instructor, implementing prompt engineering, validation, and automated extraction across multiple financial document workflows.
 - Full-Stack Development:** Developed internal AI applications using React and Streamlit, integrated backend APIs, contributed to CRM development and ServiceNow agentic workflows, and deployed solutions on Microsoft Azure infrastructure.
 - Software Engineering:** Maintained production-quality Python code following modular architecture, PEP8 standards, Git workflows, Ruff linting, modern dependency management (uv), and Agile development practices with Kanban-based sprint planning.

PROJECTS

- LexAI (Hybrid RAG, Full-Stack, Cloud)** Web | GitHub:
 - Built a Legal AI platform indexing **26,274 Supreme Court judgments** for natural-language search, precedent retrieval, and AI-assisted case analysis.
 - Engineered Hybrid RAG using **BM25 (SQLite FTS5), FAISS, PageRank, and Weighted RRF** to combine keyword, semantic, and graph-based retrieval for improved ranking quality.
 - Developed secure FastAPI REST/WebSocket APIs, optimized SQLite with WAL for concurrent access, containerized with Docker, and deployed the backend on **Google Cloud Run** with web and Flutter clients.
- PRESCRIPTION (FastAPI, Whisper, Groq)** Live | GitHub:
 - Built a voice-enabled medical assistant that converts doctor consultations into structured digital prescriptions and printable reports.
 - Integrated **Faster-Whisper** for speech recognition, LLM-based medical entity extraction, and **RapidFuzz** matching across **30,000+ Indian medicines** for reliable term identification.
 - Developed FastAPI APIs, automated PDF generation, Dockerized the application, and added **48 PyTest** unit and integration tests.
- Legal Knowledge Graph Pipeline (NLP, OCR, NetworkX)** GitHub:
 - Built an end-to-end pipeline transforming unstructured legal PDFs into structured entity-relation knowledge graphs for legal intelligence.
 - Combined PDF parsing and Tesseract OCR with **spaCy NER, RapidFuzz** entity resolution, deterministic ID mapping, and relation normalization to build consistent graph data.
 - Generated interactive graphs using **NetworkX and PyVis**, enabling semantic retrieval, downstream analytics, and exploration of relationships across legal documents.

CERTIFICATIONS

- Microsoft Certified: Azure Data Fundamentals:** Microsoft (Jun 2025)

ACHIEVEMENTS & ACTIVITIES

- Top 12 team, Mahakumbh Hackathon
- Solved 150+ Data Structures and Algorithms problems on LeetCode
- Event Lead, Linpack Club, VIT Bhopal - Organized 5+ technical workshops with 200+ participants
- Represented university volleyball team (Thunders) at Aavahan sports fest, VIT Bhopal